FULL PAPER

Procedural Content Generation in Digital Narratives

John Smith[1], **Alice Jones**[2]

[1]Department of Computer Science, State University, USA
[2]School of Digital Media, Tech Institute, UK
Corresponding: john.smith@stateuniv.edu

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Abstract

This paper demonstrates the Ludus Academik journal template within the Inkwell extension. We present computational examples that generate figures directly from markdown, combining mathematical analysis with reproducible output. The two-column layout, themed headers, and bibliography are all produced from YAML frontmatter and Pandoc compilation.

Keywords: literate programming; reproducible research; Pandoc; LaTeX

1. Introduction

Academic publishing requires precise formatting that varies by journal. Inkwell addresses this by compiling markdown to journal-specific LaTeX classes through Pandoc ([]). This document uses the Ludus Academik template, producing a two-column layout with themed section headers.

The literate programming paradigm ([]) allows code and prose to coexist. Inkwell extends this to compiled PDF output: code blocks execute, and their results (figures, tables, text) appear in the final document.

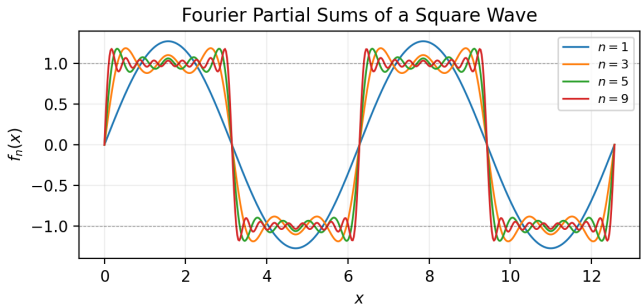

Figure 1. Fourier partial sums converging to a square wave.

2. Computational Example

We demonstrate with a Fourier series visualization ([]). The partial sum approximating a square wave is:

$$f_n(x) = \sum_{k=1}^{n} \frac{4}{(2k-1)\pi} \sin\big((2k-1)x\big)$$

3. Data Visualization

Inline Python generates a scatter plot with regression:

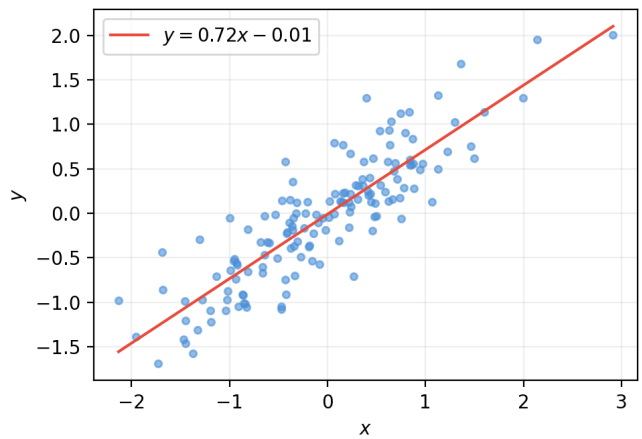

Figure 2. Simulated regression with n = 150 data points.

4. Results

Component	Status
Component	Status
Template	Functional
Code blocks	Functional
Citations	Functional
Figure captions	Functional

Table 1. Feature status for the Ludus template demo.

5. Conclusion

The Ludus template compiles correctly from markdown through Inkwell, producing the journal's native two-column layout with themed headers, figure captions, and bibliography.

References

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[] MacFarlane, John. 2023. *Pandoc: A Universal Document Converter*. Released. https://pandoc.org.

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